ML

Part V Artificial Neural Networks (ANN)

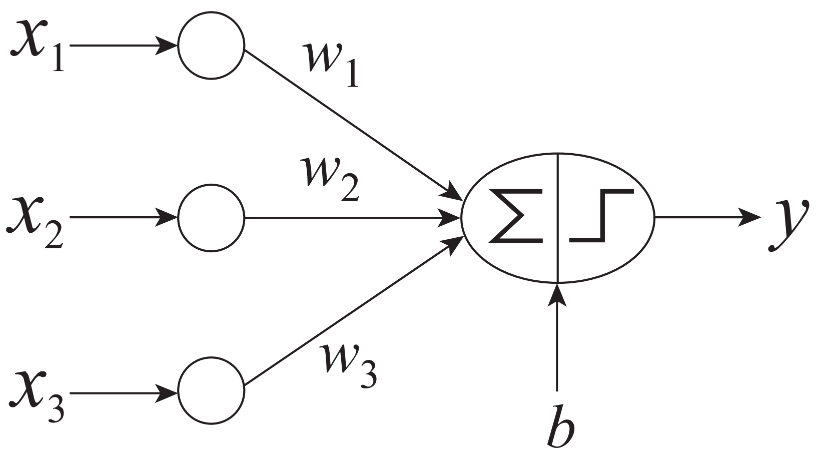
Artificial Neural Networks (ANN)

Basic Idea: A complex non-linear function can be learned as a composition of simple processing units

ANN is a collection of simple processing units (nodes) that are connected by directed links (edges)

- Every node receives signals from incoming edges, performs computations, and transmits signals to outgoing edges
- Analogous to *human brain* where nodes are neurons and signals are electrical impulses
- Weight of an edge determines the strength of connection between the nodes
- Simplest ANN: **Perceptron** (single neuron)

Basic Architecture of Perceptron



$$y = \begin{cases} 1, & \text{if } \mathbf{w}^T\mathbf{x} + b > 0. \\ -1, & \text{otherwise.} \end{cases}$$

$$\tilde{\mathbf{w}} = (\mathbf{w}^T \ b)^T \qquad \tilde{\mathbf{x}} = (\mathbf{x}^T \ 1)^T$$

$$\hat{y} = sign(\tilde{\mathbf{w}}^T\tilde{\mathbf{x}})$$

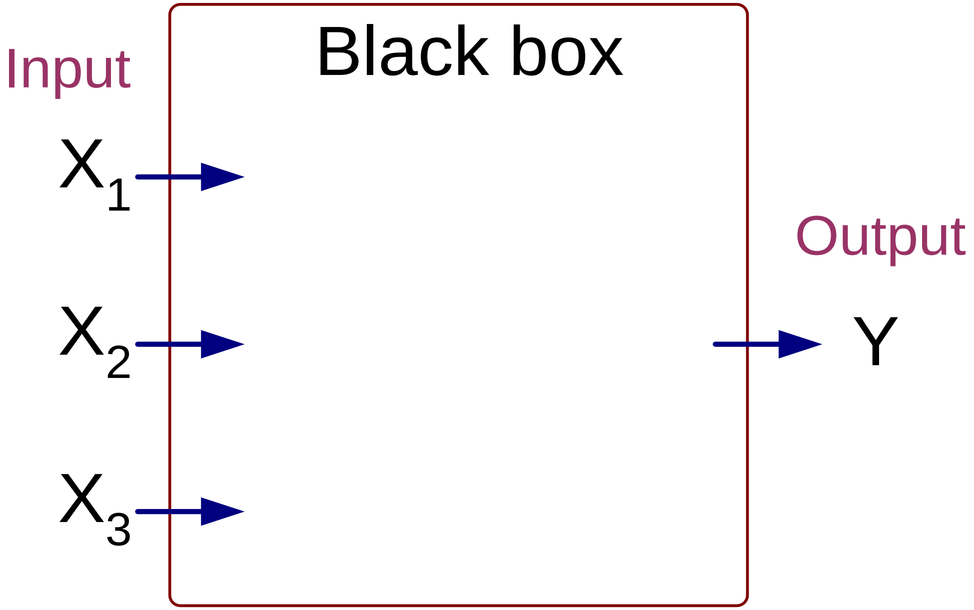
Activation Function

Learns linear decision boundaries

Similar to logistic regression (activation function is sign instead of sigmoid)

Perceptron Example

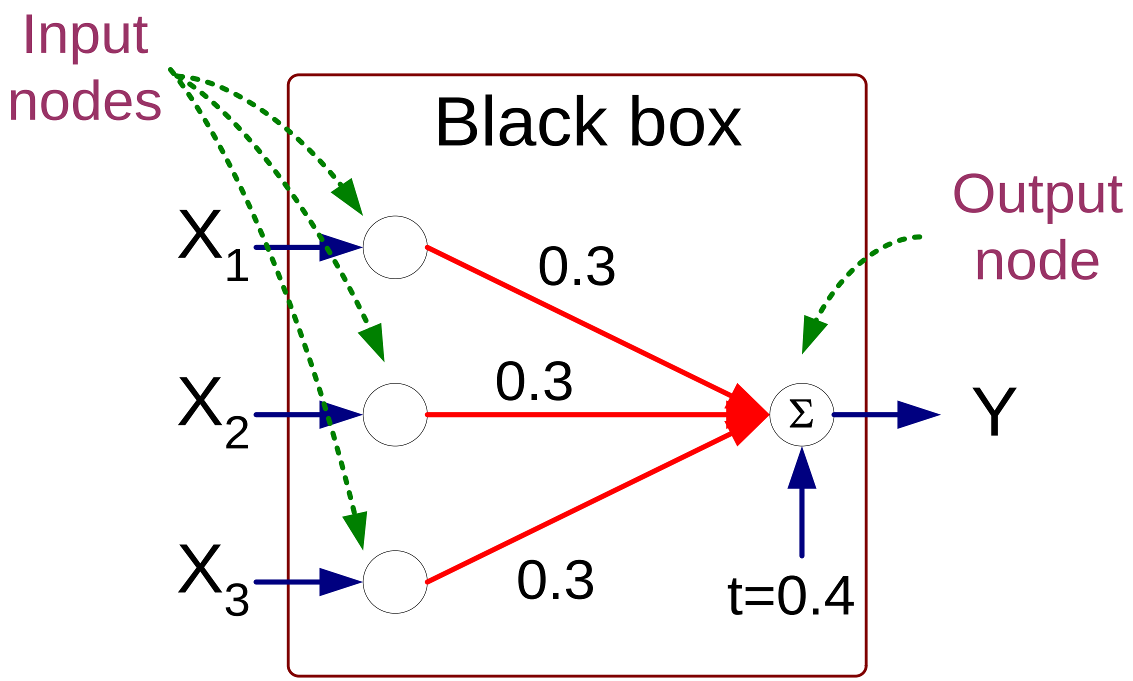
X_1	X_2	X_3	Y
1	0	0	-1
1	0	1	1
1	1	0	1
1	1	1	1
0	0	1	-1
0	1	0	-1
0	1	1	1
0	0	0	-1



Output Y is 1 if at least two of the three inputs are equal to 1.

Perceptron Example

X_1	X_2	X_3	Y
1	0	0	-1
1	0	1	1
1	1	0	1
1	1	1	1
0	0	1	-1
0	1	0	-1
0	1	1	1
0	0	0	-1



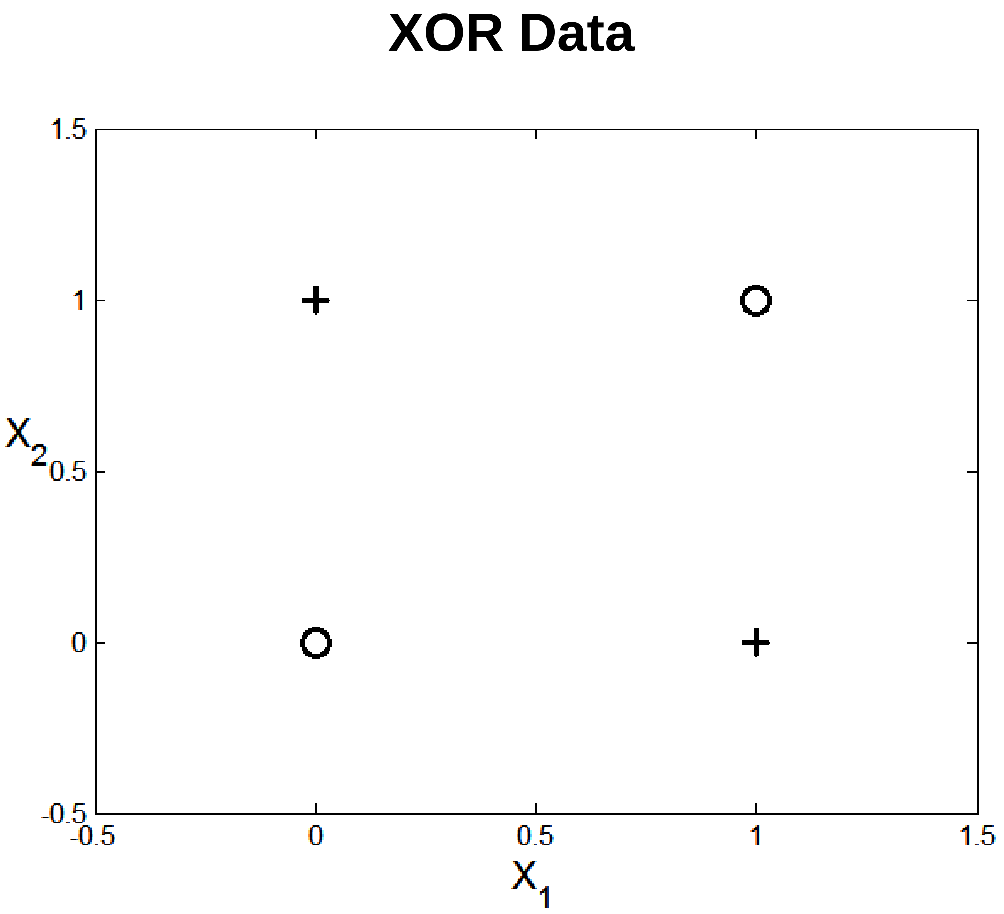
$$Y = sign(0.3X_1 + 0.3X_2 + 0.3X_3 - 0.4)$$

$$\text{where } sign(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$$

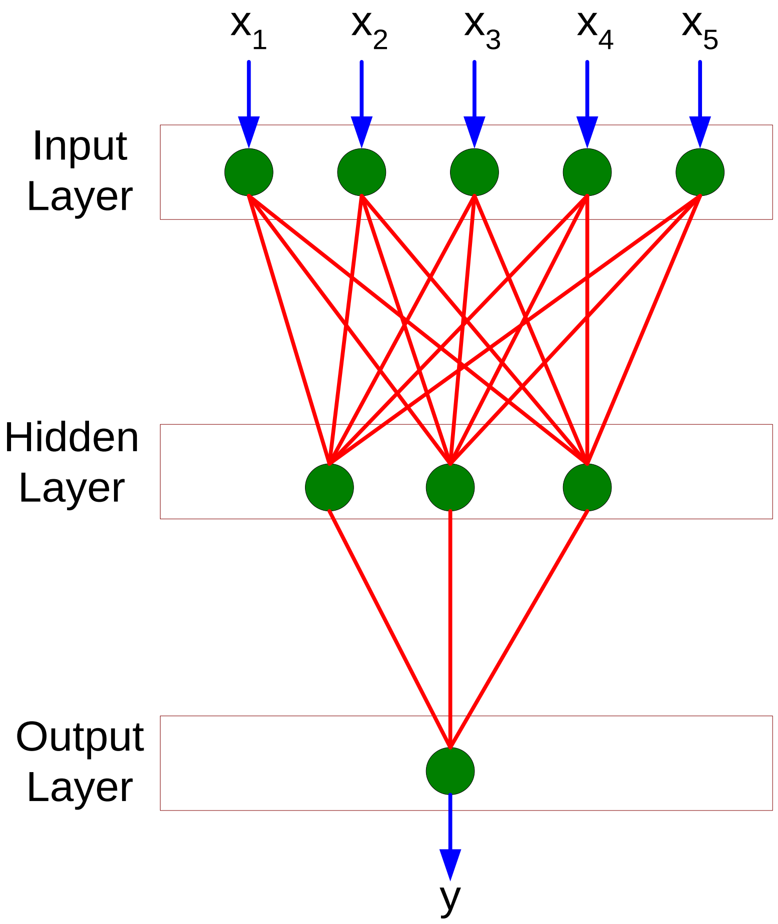
Nonlinearly Separable Data

$$y = x_1 \oplus x_2$$

x_1	x_2	y
0	0	-1
1	0	1
0	1	1
1	1	-1



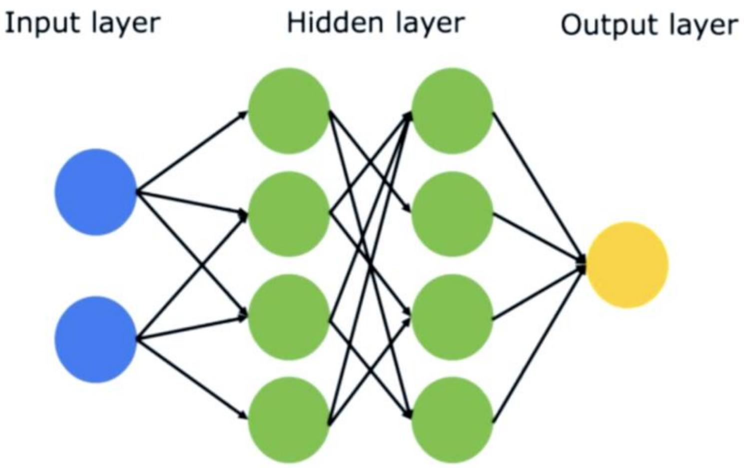
Multi-layer Neural Network



More than one *hidden layer* of computing nodes

Every node in a hidden layer operates on activations from preceding layer and transmits activations forward to nodes of next layer

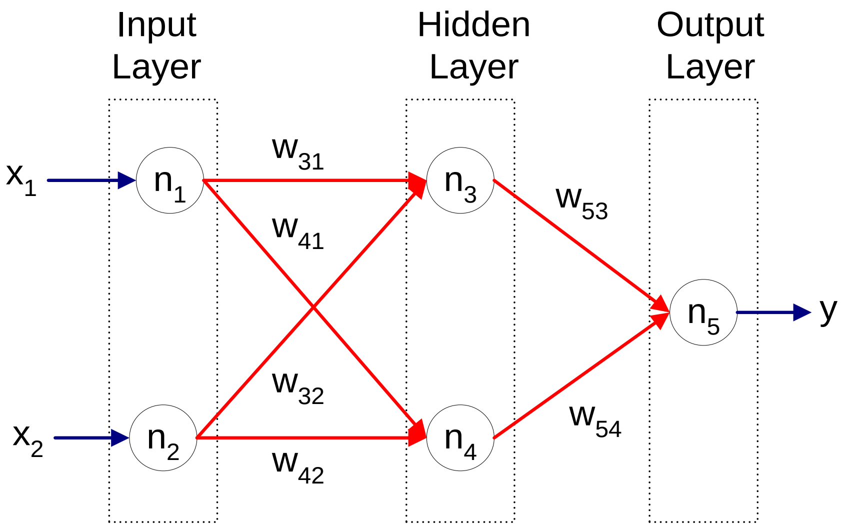
Also referred to as “feedforward neural networks”



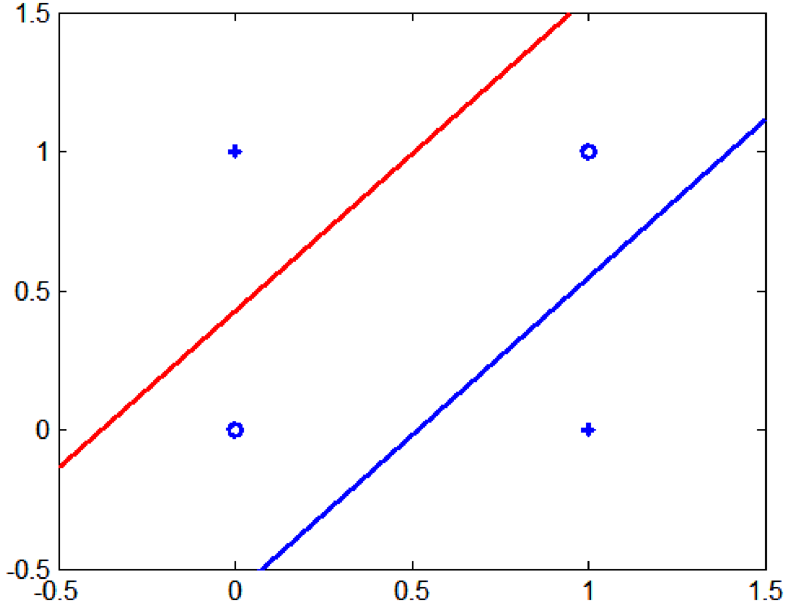
Artificial neural networks

Multi-layer Neural Network

Multi-layer neural networks with at least one hidden layer can solve any type of classification task involving nonlinear decision surfaces



XOR Data

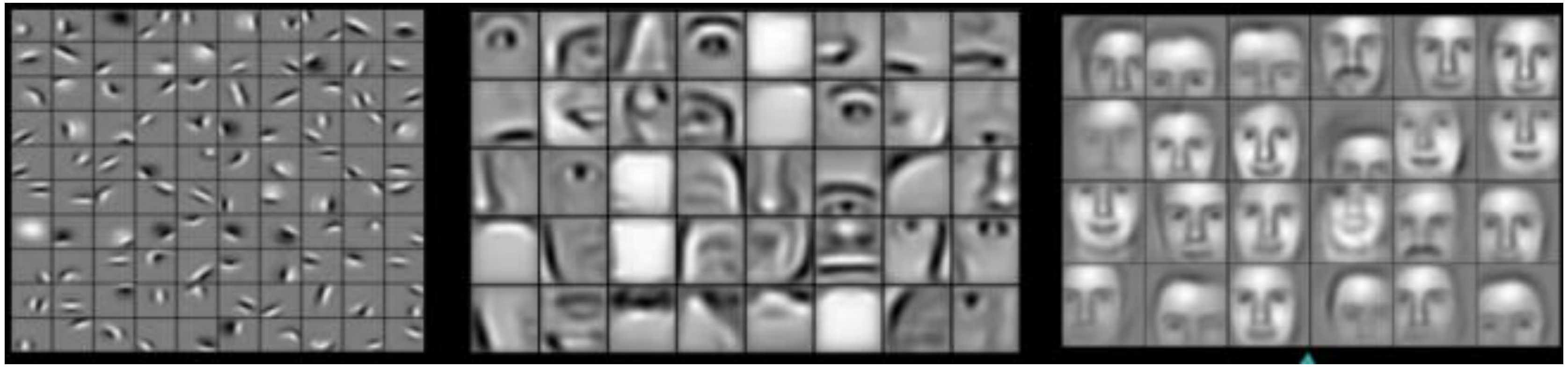


Why Multiple Hidden Layers?

Activations at hidden layers can be viewed as features extracted as functions of inputs

Every hidden layer represents a level of abstraction

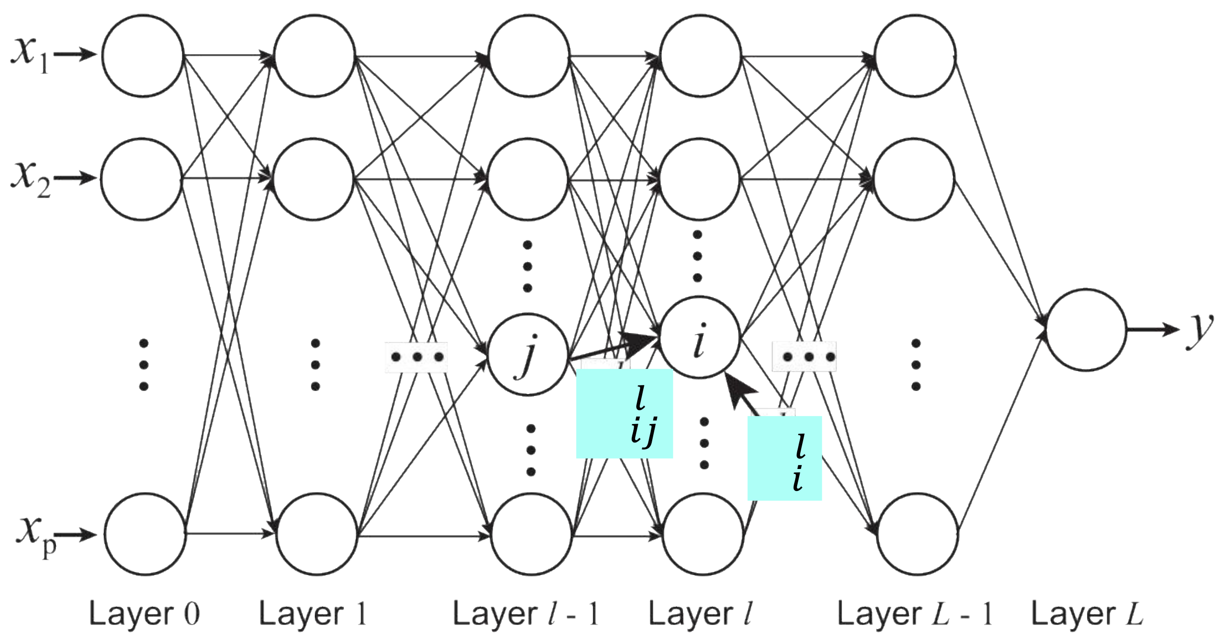
- *Complex features are compositions of simpler features*



Number of layers is known as **depth** of ANN

- *Deeper networks express complex hierarchy of features*

Multi-Layer Network Architecture



$$a_i^l = f(z_i^l) = f\Big(\underbrace{\sum_j w_{ij}^l a_j^{l-1} + b_i^l}_{\text{Linear Predictor}}\Big)$$

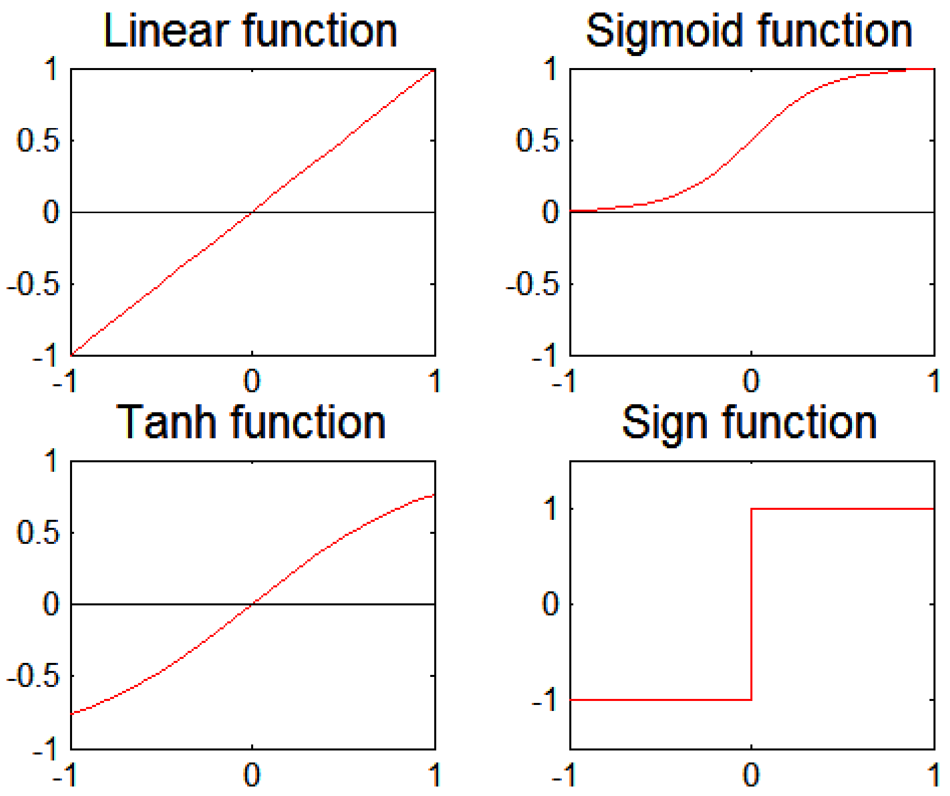
Activation value
at node i at layer l

Activation Function

Linear Predictor

Activation Functions

$$a_i^l = f(z_i^l) = f\Big(\sum_j w_{ij}^l a_j^{l-1} + b_i^l\Big)$$



$$a_i^l = \sigma(z_i^l) = \frac{1}{1+e^{-z_i^l}}.$$

$$\frac{\partial a_i^l}{\partial z_i^l} = \frac{\partial\ \sigma(z_i^l)}{\partial z_i^l} = a_i^l(1-a_i^l)$$

Learning Multi-layer Neural Network

Can we apply perceptron learning rule to each node, including hidden nodes?

- Perceptron learning rule computes error term e = y - $\hat{y}$ and updates weights accordingly
  - Problem: how to determine the true value of y for hidden nodes?
- Approximate error in hidden nodes by error in the output nodes
  - Problem:
    - Not clear how adjustment in the hidden nodes affect overall error
    - No guarantee of convergence to optimal solution

Gradient Descent

Loss Function to measure errors across all training points

$$E(\mathbf{w}, \mathbf{b}) = \sum_{k=1}^{n} \text{Loss}\ (y_k,\ \hat{y}_k)$$

Squared Loss:

$$\text{Loss}\ (y_k,\ \hat{y}_k) = (y_k - \hat{y}_k)^2.$$

Gradient descent: Update parameters in the direction of "maximum descent" in the loss function across all points

$$w_{ij}^l \longleftarrow w_{ij}^l - \lambda \frac{\partial E}{\partial w_{ij}^l},$$

$$b_i^l \longleftarrow b_i^l - \lambda \frac{\partial E}{\partial b_i^l},$$

λ: learning rate

Stochastic gradient descent (SGD): update the weight for every instance (minibatch SGD: update over min-batches of instances)

Computing Gradients

$$\frac{\partial E}{\partial w_{ij}^l} = \sum_{k=1}^{n} \frac{\partial \text{ Loss } (y_k,\ \hat{y_k})}{\partial w_{ij}^l}.$$

$$\hat{y} = a^L$$

$$a_i^l = f(z_i^l) = f\big(\sum_j w_{ij}^l a_j^{l-1} + b_i^l\big)$$

Using chain rule of differentiation (on a single instance):

$$\frac{\partial \text{ Loss}}{\partial w_{ij}^l} = \frac{\partial \text{ Loss}}{\partial a_i^l} \times \frac{\partial a_i^l}{\partial z_i^l} \times \frac{\partial z_i^l}{\partial w_{ij}^l}.$$

For sigmoid activation function:

$$\begin{aligned} \frac{\partial \text{ Loss}}{\partial w_{ij}^l} &= \delta_i^l \times a_i^l(1 - a_i^l) \times a_j^{l-1}, \\ \text{where } \delta_i^l &= \frac{\partial \text{ Loss}}{\partial a_i^l}. \end{aligned}$$

How can we compute δ_i^l for every layer?

Backpropagation Algorithm

At output layer L:

$$\delta^L = \frac{\partial \text{ Loss}}{\partial a^L} = \frac{\partial \ (y - a^L)^2}{\partial a^L} = 2(a^L - y).$$

At a hidden layer l (using chain rule):

$$\delta_j^l \ = \ \sum_i (\delta_i^{l+1} \times a_i^{l+1}(1 - a_i^{l+1}) \times w_{ij}^{l+1}).$$

- Gradients at layer l can be computed using gradients at layer l + 1
- Start from layer L and “backpropagate” gradients to all previous layers

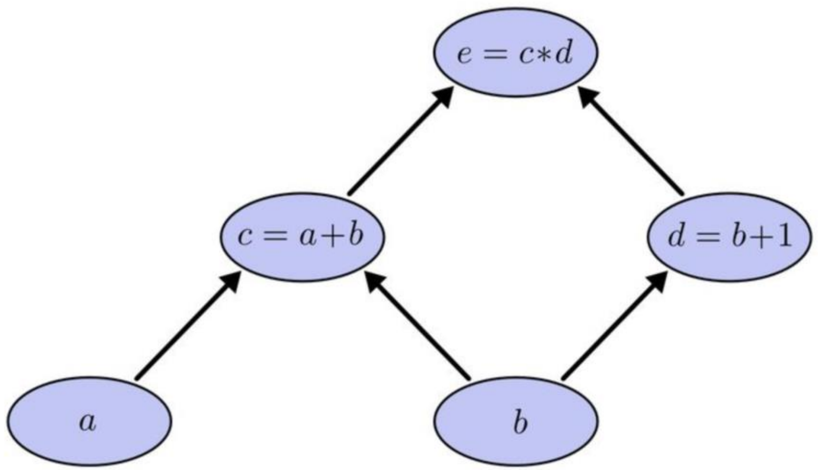
Use gradient descent to update weights at every epoch

For next epoch, use updated weights to compute loss fn. and its gradient

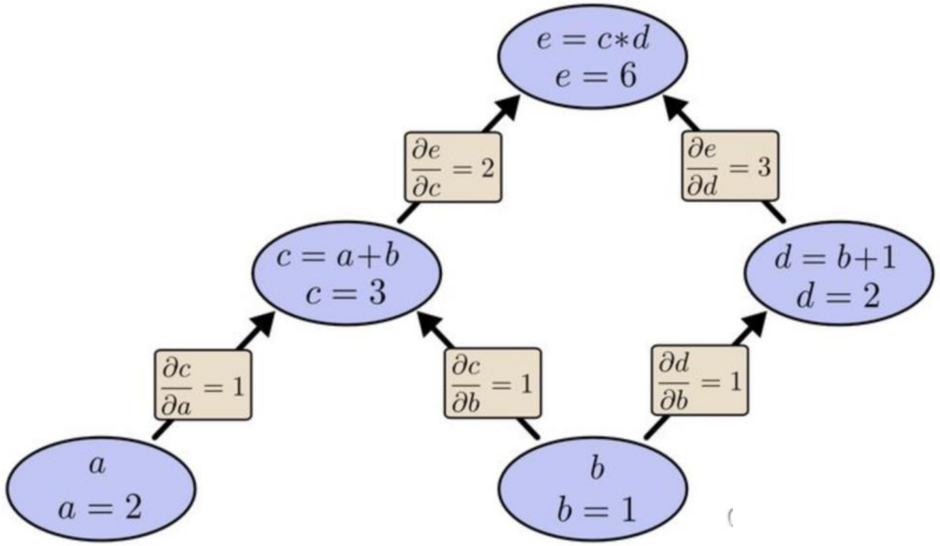
Iterate until convergence (loss does not change)

Chain Rule of Derivatives

Chain Rule Example



1. $\frac{\partial e}{\partial a} = \frac{\partial e}{\partial c}\frac{\partial c}{\partial a} = 1 * 2$
2. $\frac{\partial e}{\partial b} = \frac{\partial e}{\partial c}\frac{\partial c}{\partial b} + \frac{\partial e}{\partial d}\frac{\partial d}{\partial b} = 2 * 1 + 3 * 1 = 5$



求偏导可以一直累乘到叶子节点，并求和

Design Issues in ANN

- Number of nodes in input layer
  - One input node per binary/continuous attribute
  - k or $\log_2$ k nodes for each categorical attribute with k values
- Number of nodes in output layer
  - One output for binary class problem
  - k or $\log_2$ k nodes for k-class problem
- Number of hidden layers and nodes per layer
- Initial weights and biases
- Learning rate, max. number of epochs, mini-batch size for mini-batch SGD, …

Characteristics of ANN

- Multilayer ANN are universal approximators but could suffer from overfitting if the network is too large
- Gradient descent may converge to local minimum
- Model building can be very time consuming, but testing can be very fast
- Can handle redundant and irrelevant attributes because weights are automatically learnt for all attributes
- Sensitive to noise in training data
- Difficult to handle missing attributes

Deep Learning Trends

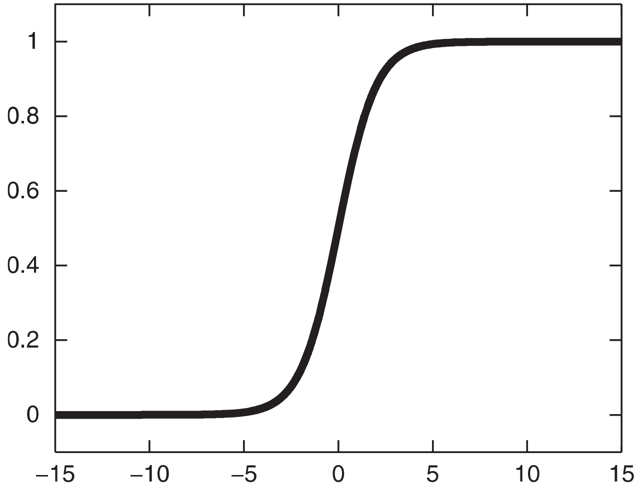
Training **deep** neural networks (more than 5-10 layers) could only be possible in recent times with:

- Faster computing resources (GPU)
- Larger labeled training sets
- **Algorithmic Improvements in Deep Learning**

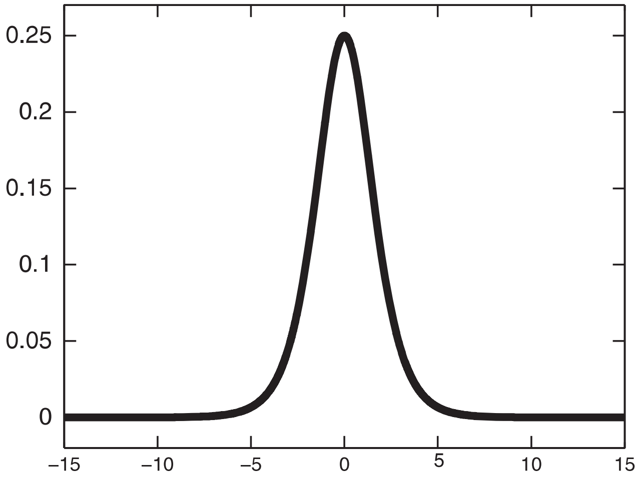
Recent Trends:

- Specialized ANN Architectures:
  - Convolutional Neural Networks (for image data)
  - Recurrent Neural Networks (for sequence data)
  - Residual Networks (with skip connections)
- Unsupervised Models: Autoencoders
- Generative Models: Generative Adversarial Networks

Vanishing Gradient Problem



(a) $\sigma(z)$.



(b) $\partial\sigma(z)/\partial z$.

Sigmoid activation function easily saturates (show zero gradient with z) when z is too large or too small

Lead to small (or zero) gradients of squared loss with weights, especially at hidden layers, leading to slow (or no) learning

$$\frac{\partial \text{ Loss}}{\partial w_j^L} = 2(a^L - y) \times \sigma(z^L)(1 - \sigma(z^L)) \times a_j^{L-1}.$$

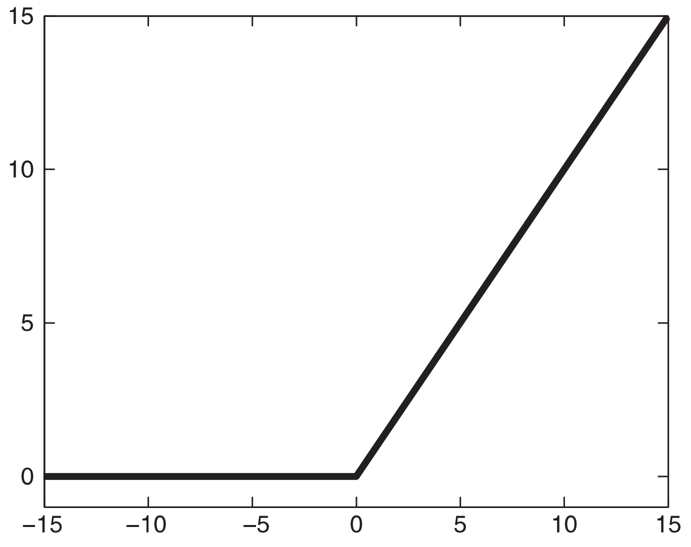
Handling Vanishing Gradient Problem

Use of Cross-entropy loss function

$$Loss\ (y,\ \hat{y}) = -y\log(\hat{y}) - (1-y)\log(1-\hat{y})$$

$$\frac{\partial\ \text{Loss}}{\partial w_j^L} = (a^L - y) \times a_j^{L-1}$$

Use of Rectified Linear Unit (ReLU) Activations:



$$a = f(z) = \begin{cases} z, & \text{if } z > 0. \\ 0, & \text{otherwise.} \end{cases}$$

$$\frac{\partial a}{\partial z} = \begin{cases} 1, & \text{if } z > 0. \\ 0, & \text{if } z < 0. \end{cases}$$

Try something out